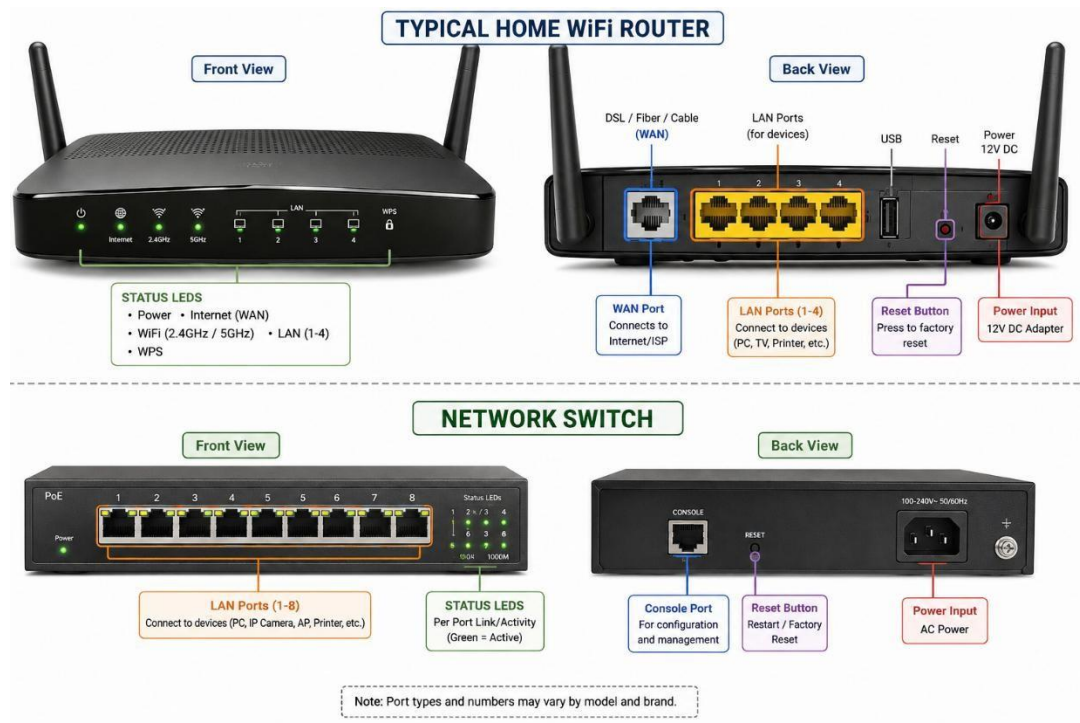


1. Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.

ANS:-



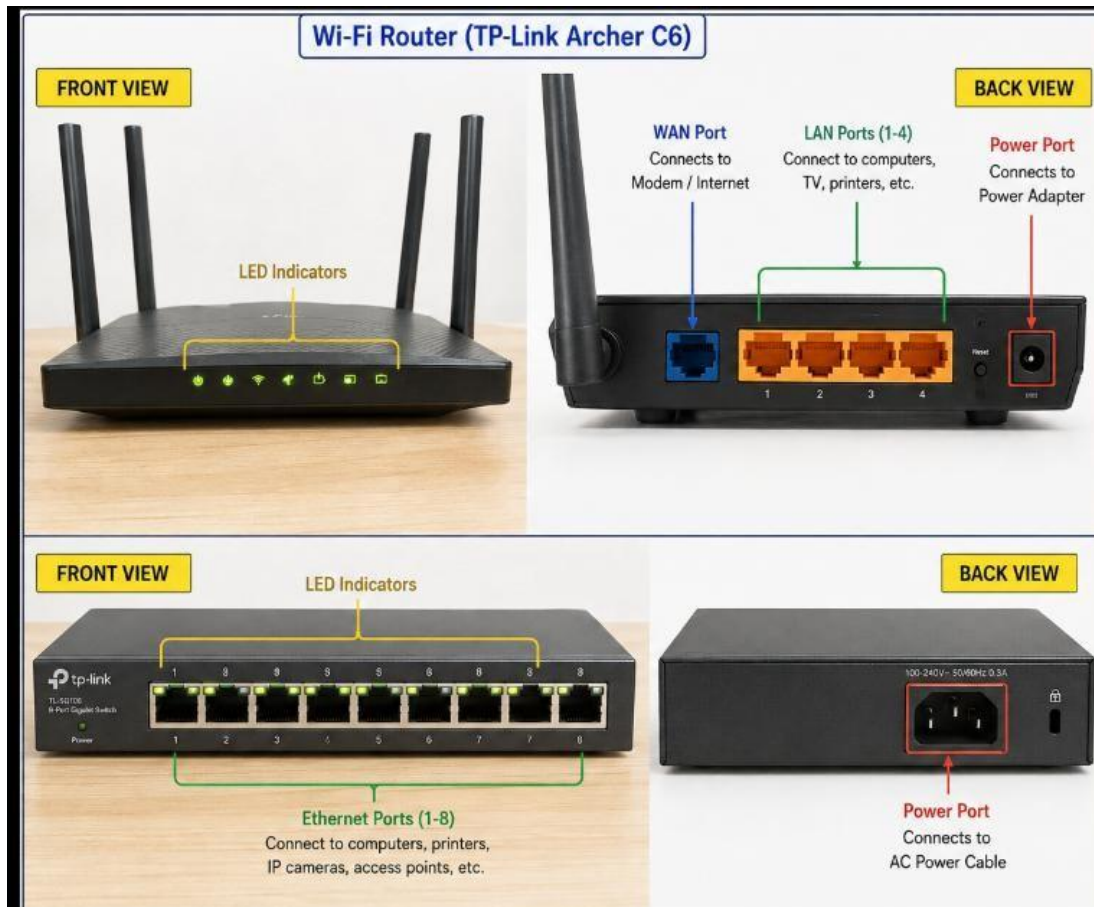
2. List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic. Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

ANS:-

Point	Router	Switch
1. Main Function	Connects different networks , such as a home network to the Internet.	Connects devices within the same local network (LAN) .
2. Address Used	Mainly uses IP addresses to decide where to forward packets.	Mainly uses MAC addresses to forward Ethernet frames to the correct device.
3. Traffic & Networks	Routes traffic between different networks and can connect a LAN to the Internet/WAN.	Sends traffic between devices on the same LAN , such as PCs, printers, and servers.

3. Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.

ANS:-



Wi-Fi Router – Key Points

1. Power Port (Back)

- Supplies power to the router using the adapter.

2. Ethernet Ports

- **WAN Port (Blue):** Connects to modem/internet.
- **LAN Ports (Orange):** Connect to local devices (PC, TV, etc.)

3. LED Indicators (Front)

- Power – Shows if router is ON.
- Internet – Shows internet connection status.
- Wi-Fi – Shows Wi-Fi is active.
- LAN (1-4) – Shows connection/activity on each LAN port.
- WPS – Shows WPS pairing status.

Network Switch – Key Points

1. Power Port (Back)

- Connects to AC power cable to turn on the switch.

2. Ethernet Ports (Front)

- Ports 1 to 8 connect network devices in the same local network.

3. LED Indicators (Front)

- Green LED above each port:
 - ON = Link/Connection established • Blinking = Data is being transmitted
- Power LED:
 - ON = Switch is powered on

4. Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

ANS:-

Small Gaming Café Network Scenario

For a small gaming café, I would connect the **router first to the ISP line**. The router receives the Internet connection from the ISP and manages the network using IP addresses.

Next, I would connect the router's **LAN port to a network switch** using an Ethernet cable. Then, I would connect the **8 gaming PCs to 8 Ethernet ports on the switch**.

Connection:

ISP → Router → Switch → 8 Gaming PCs

The **router** is needed to provide Internet access for IPL streaming and online multiplayer games, while the **switch** allows all 8 gaming PCs to communicate efficiently over the local network. Using a switch also provides dedicated Ethernet connections, which can give better stability and lower latency than relying on Wi-Fi for gaming.